HDFS Configuration Single Cluster

core-site.xml

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>

  <property>
    <name>hadoop.permissions.enabled</name>
    <value>false</value>
  </property>
</configuration>
```

yarn-site.xml

```
<configuration>
  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>

  <property>
    <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
```

```xml
    <value>org.apache.hadoop.mapred.ShuffleHandler</value>
  </property>


  <property>
    <name>yarn.nodemanager.local-dirs</name>
    <value>C:/hadoop-3.3.6/tmp/nm-local-dir</value>
  </property>


<property>
  <name>yarn.nodemanager.local-dirs-permission-check.enabled</name>
  <value>false</value>
</property>


 <property>
  <name>yarn.nodemanager.log-dirs</name>
  <value>C:/hadoop-3.3.6/tmp/logs</value>
</property>

  <property>
    <name>yarn.nodemanager.resource.memory-mb</name>
    <value>4096</value>
  </property>

  <property>
    <name>yarn.nodemanager.resource.cpu-vcores</name>
    <value>2</value>
  </property>

  <property>
    <name>yarn.scheduler.maximum-allocation-mb</name>
```

```
   <value>4096</value>
  </property>

  <property>
   <name>yarn.scheduler.maximum-allocation-vcores</name>
   <value>2</value>
  </property>

</configuration>
```

mapred-site.xml

```
<configuration>
 <property>
   <name>mapreduce.framework.name</name>
   <value>yarn</value>
  </property>
</configuration>
```

hdfs-site.xml

```
<configuration>
 <property>
   <name>dfs.replication</name>
   <value>1</value>
 </property>
 <property>
   <name>dfs.namenode.name.dir</name>
   <value>C:\hadoop-3.3.6\data\namenode</value>
 </property>
 <property>
   <name>dfs.datanode.data.dir</name>
   <value>C:\hadoop-3.3.6\data\datanode</value>
 </property>
<property>
 <name>dfs.permissions</name>
 <value>false</value>
</property>

</configuration>
```

Hadoop Environment Setup Notes (Windows)

- **JAVA_HOME**
  - It tells Windows where Java is installed.
  - **Example:**
    - C:\Program Files\Java\jdk1.8.0_202
- **HADOOP_HOME**
  - Path where Hadoop is installed.
  - **Example:**
    - C:\Users\PERCOIDIT\Desktop\hadoop-3.3.6
- **PATH Variable**
  - It allows commands like java, hadoop, hdfs to run from anywhere.
  - Click **Path** → **Edit** → **New**
    - %JAVA_HOME%\bin
    - %HADOOP_HOME%\bin
    - %HADOOP_HOME%\sbin

//Note:

- **Data Folder Creation**
  - For this we need to create one data folder inside hadoop folder and inside data folder we need to create two folders named as **datanode** and **namenode.** After that we need to provide those two paths respectively in hdfs-site.xml file
- **WindowsUtils for Hadoop**
  - After downloading the winutils folder as well, copy the hadoop.dll and winutils file to the **bin** folder of hadoop3.3.6

Native winutils : we need to delete the current bin folder of hadoop and replace with the bin from the link with proper version of hadoop

Hadoop Support

This is only needed if we want to install both version of JDK
Java 8 or 11 is needed for hdfs namenode -format command.

CommandLine

- Format the NameNode
  - Open **Command Prompt as Administrator** and run:
    - hdfs namenode -format
- Start Hadoop[go to the sbin folder of the hadoop]
  - start-dfs.cmd
  - Start-yarn.cmd

Modern alternative to start YARN and HDFS [Linux]

- **HDFS services:**
  - hdfs --daemon start namenode
  - hdfs --daemon start datanode
  - hdfs --daemon start secondarynamenode
- **YARN services:**
  - yarn --daemon start resourcemanager
  - yarn --daemon start nodemanager

- Verfify Services
  - jps
- List files in HDFS root
  - Shows all folders/files in HDFS root
    - hdfs dfs -ls /
- Create directory in HDFS
  - Creates folder /BigData in HDFS
    - hdfs dfs -mkdir /BigData
- Upload file from Windows → HDFS
  - Copies local file into HDFS
    - hdfs dfs -put C:\Users\PERCOIDIT\Desktop\B_Data\players.txt /BigData

Create Text file more than 128MB for multiple blocks

fsutil file createnew C:\Users\PERCOIDIT\Desktop\B_Data\file300.txt 300000000

hdfs dfs -put C:\Users\PERCOIDIT\Desktop\B_Data\file300.txt /BigData

Browse HDFS file

hdfs dfs -ls /

Then open these URLs in your browser:

- **HDFS:** http://localhost:9870
- **YARN:** http://localhost:8088

To run Map Reduce

hadoop jar
C:\Users\PERCOIDIT\Desktop\hadoop-3.3.6\share\hadoop\tools\lib\hadoop-strea
ming-3.3.6.jar ^
-input /WorldCup/Players.txt ^
-output /CountResult ^
-mapper " python C:\Users\PERCOIDIT\Desktop\B_Data\mapper.py" ^
-reducer "python C:\Users\PERCOIDIT\Desktop\B_Data\reducer.py"

To run using Builtin Hadoop Methods

hadoop jar
%HADOOP_HOME%\share\hadoop\mapreduce\hadoop-mapreduce-examples-3.3
.6.jar wordcount /wordcount /output

View Inbuilt Methods
hadoop jar
%HADOOP_HOME%\share\hadoop\mapreduce\hadoop-mapreduce-examples-3.3
.6.jar wordcount /WorldCup/Players.txt /outputbuilt

hdfs dfs -rm -r /CountResult
hdfs dfs -cat /CountResult/part-00000